

Project Proposal

I selected the Medical Insurance Forecast dataset that comes from Medical Cost Personal Datasets on Kaggle.

Key characteristics of the dataset:

The dataset is stored as a CSV file containing 1338 records and seven columns. Of the six predictors, age, BMI, and number of children are numerical, while sex, smoking status, and region are categorical. The remaining column, charges, is the numerical target and represents the medical costs we aim to predict.

Data cleaning:

Firstly, I will check that all required columns are present, look for missing values, and remove exact duplicate records.

Secondly, I will make sure the numerical columns are stored as numbers and check for invalid values, such as negative ages or children counts that are not whole numbers, while keeping valid high charges.

Thirdly, I will make the categorical labels consistent by removing extra spaces and using the same spelling and capitalization, then represent categories using columns containing 0 or 1, such as 1 for smokers and 0 for nonsmokers, so the regression model can use them.

Initial visualizations:

Firstly, I will use a histogram to examine the distribution of medical charges and identify which charge ranges occur most often.

Secondly, I will use a box plot to compare the median medical charges and how much charges vary within the smoker and nonsmoker groups.

Thirdly, I will use scatter plots of age and BMI against charges, with points coloured by smoking status, to examine how age and BMI are related to charges and whether these patterns differ between the two groups.

Regression model:

I will use multiple linear regression to predict medical charges using age, BMI, number of children, sex, smoking status, and region. This model is appropriate because medical charges are continuous numerical values, and it allows several predictors to be considered together. I will also compare the basic multiple linear regression model with another multiple linear regression model that allows different relationships between BMI and medical charges for smokers and nonsmokers.